Response to HHS Health Sector AI RFI (RIN 0955-AA13)

Translational Physical AI Platform for Medical Clinical Care Delivery, Education and Training.

Submitted by: MediView XR, Inc.

Executive Summary

MediView XR respectfully submits this response to the HHS Request for Information regarding accelerating the adoption and use of Artificial Intelligence (AI) as part of clinical care. MediView represents a translational application of 'Physical AI'—the integration of artificial intelligence, real-time 3D medical data, and spatial computing directly into the physical act of care delivery.

Unlike retrospective or purely analytical AI tools, MediView's platform operates in situ during live clinical procedures, translating multimodal imaging and sensor data into spatially aligned, actionable guidance. The system closes the loop between data → cognition → physical execution. This approach supports clinician education, telemedicine, procedural guidance, trauma care, and other high-acuity interventions.

I. Defining Physical AI in Clinical Care

Most healthcare AI systems today focus on image interpretation, predictive analytics, risk scoring, or retrospective decision support. While valuable, these tools often stop at generating insight, requiring clinicians to translate outputs into physical action.

Physical AI differs in that it operates in real time, maintains spatial awareness of anatomy and instruments, preserves human-in-the-loop control, and directly influences procedural execution and outcomes. MediView's mixed reality platform fuses CT, MRI, live ultrasound, and tracking data into AI-assisted 3D anatomical reconstructions that are spatially registered to the patient during care and remotely connected for collaborative care delivery.

II. Technical Architecture and Embedded AI

MediView XR functions as a real-time spatial computing platform integrating:

- Pre-operative and intra-operative imaging (CT, MRI, ultrasound)
- AI-enabled segmentation and image registration
- Continuous spatial alignment and instrument tracking
- Mixed Reality visualization via head-mounted displays

AI processes include automated anatomical segmentation, volumetric reconstruction, multi-modal image fusion, and dynamic spatial registration. These processes are embedded within the visualization layer rather than delivered as separate analytic outputs, transforming static imaging into dynamic, patient-specific procedural guidance. In essence, **the system translates the disparate data inputs from existing and emerging technologies (building upon other investments and efforts) into real-world visual 3D representations, tracking and guidance for imaging, diagnostics and procedural delivery**.

This spatial intelligence reduces cognitive load, improves hand-eye coordination, enhances anatomical comprehension, and increases procedural confidence—particularly during high-acuity or time-sensitive interventions. Moreover, with connectivity the system can allow practitioners to remotely collaborate for physical care delivery across rural and underserved or understaffed locations.

III. Clinical Applications of Physical AI

Education and Training:

Physical AI enables 3D anatomy-accurate training and procedural rehearsal using patient-specific data. Mixed Reality environments allow clinicians to understand spatial relationships rather than memorizing steps, accelerating skill acquisition and improving retention.

Telemedicine and Remote Expert Guidance:

Shared holographic visualization allows remote experts to observe precisely what frontline clinicians see, facilitating real-time procedural guidance with spatial fidelity. This expands access to subspecialty expertise and reduces unnecessary patient transfers.

From Trauma and Emergency Care to Complex Procedures:

In acute and resource-constrained environments, Physical AI supports rapid localization of internal anatomy and enables image-guided interventions without reliance on multiple external monitors or complex imaging workstations.

IV. Targeted Responses to Specific RFI Questions

1. Barriers to Innovation and Adoption:

Key barriers include the need for investment for integration of varying technologies into the mixed reality ecosystem and related collaboration.

2. Regulatory and Payment Policy Priorities:

HHS can accelerate adoption through targeted investment into the foundational platform and by clarifying predictable regulatory pathways for Physical AI platforms that blend medical devices, Software as a Medical Device (SaMD), and AI-enabled workflows. Lifecycle evaluation models for adaptive systems should be encouraged. Reimbursement reforms should explore bundled or value-based models recognizing improved first-pass success, reduced complications, and downstream cost avoidance.

3. Legal and Implementation Considerations:

Human-in-the-loop AI systems require clear governance structures around liability, transparency, data security, and accountability. HHS can support guidance clarifying roles and responsibilities for clinicians and technology providers in AI-supported care environments.

4. Evaluation Methods and Metrics:

Promising evaluation methods include prospective clinical trials, real-world evidence studies, human factors testing, cognitive load measurement, robustness testing under workflow variability, and post-deployment performance monitoring.

5. Supporting Private Sector Activities:

HHS can encourage accreditation programs, industry-driven benchmarking, certification pathways, and public-private collaborations to standardize evaluation metrics for Physical AI systems.

6. High-Impact AI Use Cases:

AI tools that directly enhance procedural accuracy, reduce complication rates, shorten time to intervention, and improve system efficiency hold tremendous potential for improving outcomes, lowering costs and democratizing care across populations.

7. Organizational Adoption Dynamics:

Clinical leadership, hospital administrators, IT governance bodies, and reimbursement decision-makers significantly influence AI adoption. Administrative hurdles include procurement complexity, integration challenges, and uncertainty regarding reimbursement.

8. Interoperability Needs:

Enhanced interoperability for 3D volumetric imaging, spatial metadata, procedural data standards, and AI model integration would accelerate innovation and widen market opportunities.

9. Patient and Caregiver Perspectives:

Patients seek improved accuracy, reduced complications, shorter procedures, and expanded access to expertise. Concerns include transparency, data privacy, safety, and preservation of clinician oversight.

10. Research & Investment Priorities:

HHS should prioritize investment in the foundational platform and applied clinical research evaluating Physical AI's impact on outcomes, workflow efficiency, human factors, and costeffectiveness. Public-private partnerships can accelerate evidence generation and responsible adoption.

V. Metrics Framework

Evaluation domains include:

- Clinical performance: accuracy, complication reduction, success rates
- Operational efficiency: time to intervention, procedural duration
- Human factors: cognitive load, fatigue, ergonomics
- System-level impact: cost avoidance, resource utilization, training efficiency
- Expansion of higher-level care delivery capabilities to remote and underserved populations.

(Quantitative metrics will continue to mature as deployment expands.)

Conclusion

Physical AI represents the next evolution of healthcare AI by embedding intelligence directly into the physical act of care delivery. With targeted investment and sustained research, platforms such as MediView XR can accelerate AI adoption, improve outcomes, reduce system burden, and strengthen public trust in advanced healthcare technologies.

Respectfully submitted,

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